Multi-Perspective Stance Detection Task

HLT Report, Group 12

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1. Introduction

Large language models (LLMs) have revolutionized natural language processing (NLP) field by outperforming state-of-the-art approaches in NLP tasks. In the context of supervised learning settings, which require labeled data, it is well-known that human annotators may provide diverse perspectives via annotations due to the intricacies of thought, varied life experiences, and diverse educational backgrounds of annotators [1,2]. This inevitably leads to disagreement especially in the subjective tasks such as toxicity, argumentation mining, stance and hate speech detection in which multiple perspectives may be equally valid, and a unique 'ground truth' label may not exist. To validate this assumption, we intend to revisit the task of stance detection through involving multiple annotations, i.e. diverse perspectives of multiple annotators, instead of a single ground truth. In this study, our contributions are two-fold: (i) we investigate if designing a more inclusive ethically-aware LLM can give better classification performance, and (ii) further explore if the model confidence is affected by annotator disagreement.

The main concern related to the traditional aggregation procedures, which reduce multiple annotations into a single label using standard approaches of disagreement resolution through majority voting, is oversimplifying the real-world complications by assuming the presence of a single ground truth [3,4,5]. Due to these concerns, a novel approach called *perspectivism* has recently been proposed for gathering annotated data in NLP [6]. This new paradigm promotes the use of disaggregated datasets, containing different human judgments, especially for subjective tasks. This approach holds the potential of fostering the development of *responsible* and *ethical* AI systems in which fairness is promoted through involving diverse annotators' backgrounds (perspectives).

2. Background

What has been traditionally regarded as disagreement in the NLP community is now being recognized as human label variation (HLV), as suggested by [7]. Acknowledging that disagreement in human labeling affects every aspect of the NLP pipeline —data, modeling, and evaluation—recent studies have adopted an opposite approach [8,9,10,11,12]. These studies challenge the assumption that texts have a single interpretation and question the notion that annotator disagreement is erroneous. A survey by [13] emphasizes that relying on a gold standard as an objective truth is inappropriate for many NLP applications like machine translation and natural language generation, where human creativity is crucial. Additionally, the idea of language objectivity is often oversimplified not considering factors like word sense variations, semantic role labeling, and diverse

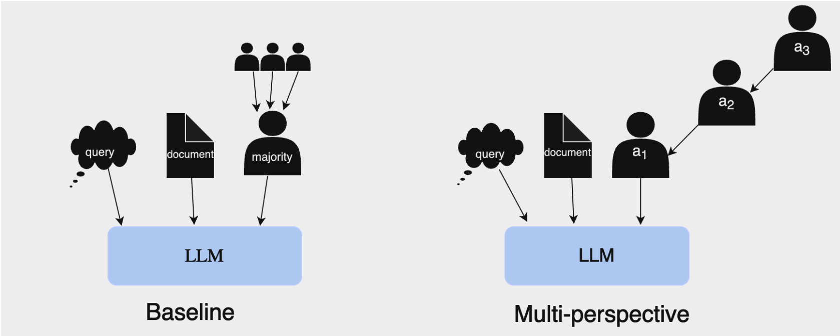


Figure 1. Baseline vs. Multi-perspective Approach in Model Finetuning

cultural backgrounds that influence humans understanding. Furthermore, [7] points out that disagreement also stems from factors such as the complexity or ambiguity of the task [14] and subjective interpretations influenced by annotators' demographics [15].

3. Methodology

In the following section we describe our methodology to employ multi-perspective stance classification.

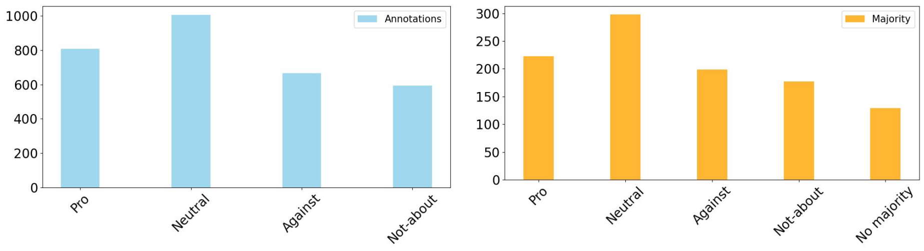
3.1. Dataset

We use the dataset collected by [16] to apply multi-perspectivist approach to stance classification task. The dataset is composed of top-10 news documents that have been crawled from the two popular search engines by issuing 57 queries in total which correspond to a wide range of controversial topics; including but not limited to education, health, entertainment, religion and politics. After the crawling, the authors further obtained the stance label of each document with respect to the queries via crowdsourcing using the MTurk[1] and each document was judged by three crowd-workers [16]. A stance label can have the following values: *pro, neutral, against, not-about,* and *link not-working*. Note that due to the nature of the MTurk platform, although there are only three annotations for each document, this does not guarantee that these annotations come from the same three annotators. Thus, there is a high probability that the annotated dataset contains more than three diverse perspectives which highly affects the disagreement level. Both the reported Fleiss-Kappa score of 0.3500 which is fair agreement, and inter-rater agreement score of 0.4968 in the original paper confirm the subjective nature and difficulty of the stance classification task.

3.2. Stance Classification

To examine the effect of multi-perspectivist approach on classification model performance, we investigate two different paradigms of including annotations to the model training phase, as *Baseline* and *Multi-perspective* models.

[1] Amazon Mechanical Turk



(a) Original label distribution

(b) Majority label distribution

Baseline Model In the baseline model, we use the traditional label aggregation schema of *majority voting*, thus there is only one majority label per document. Based on this, in the baseline dataset, each document d_i is composed of query, document content, and majority label defined as $d_i = \{q_i, c_i, m_i\}$. Then, this dataset is used as the training dataset for fine-tuning the baseline model.

Multi-perspective Model Unlike, for the multi-perspective model, we create the dataset by data augmentation through introducing more than one single label for each document. Thus, the multi-perspective dataset consists of the documents d_i, where d_i is added to the dataset three times with the corresponding annotations as ${d_i}^1 = \{q_i, c_i, a_1\}$, ${d_i}^2 = \{q_i, c_i, a_2\}$, and ${d_i}^3 = \{q_i, c_i, a_3\}$, where the annotation set of d_i is defined as $A(d_i) = \{a_1, a_2, a_3\}$ and ${a_i}^1$, ${a_i}^2$, ${a_i}^3$ do not have to be distinct. In this study, we followed the document duplication approach to prevent overfitting due to the small dataset size. Then, this augmented dataset is used as the training dataset for fine-tuning the multi-perspective model. Note that for both the baseline and multi-perspective models, model evaluation is fulfilled using the majority labels.

Chunking Since the transformer-based models have maximum length limitation, which is 512 for the base models of BERT and RoBERTa, the content is truncated to 512, if there is no chunking. In the case of chunking which is a widely used technique for handling long inputs, the input is splitted into segments (chunks) using the library of chunkipy[2]. The chunkipy library ensures to provide complete and syntactically correct sentences through using the stanza library [17] with a flexible overlapping to preserve the context along the chunks. Note that the chunks might have different lengths.

4. Experimental Setup

This section provides the description of our experimental setup based on the proposed methodology.

4.1. Dataset Preprocessing

The dataset consists of documents with varying lengths and some of them are extremely long[3]. Thus, we checked the document lengths and set the maximum length as 8,000

[2]https://pypi.org/project/chunkipy/

[3]Probably due to scraping errors, 54 documents contain 4 million tokens since when we manually checked their URLs, they are currently unavailable.

Approach	Model	Chunking	Acc.	Prec.	Rec.	F1	Avg. Conf.
Baseline	BERT-base	no	28.66	27.59	22.42	17.17	0.33
		yes	33.12	30.70	28.17	26.67	0.44
	RoBERTa-base	no	36.30	34.99	31.82	27.07	0.39
		yes	45.85	39.47	43.13	**40.48**	0.52
	Mistral 7B	no	28.77	29.36	33.73	28.46	0.54
	Llama 2 7B	no	30.21	33.74	29.42	29.07	**0.66**
Multi-Perspective	BERT-base	no	32.48	31.12	28.22	24.81	0.51
		yes	47.48	53.90	49.86	**50.21**	0.52
	RoBERTa-base	no	47.77	44.27	43.63	41.43	**0.55**
		yes	47.48	52.68	50.14	47.45	0.54
	Mistral 7B	no	39.56	43.98	38.84	38.03	0.58
	Llama 2 7B	no	30.93	32.44	28.90	27.75	**0.66**

Table 1. Overall model evaluation results for the baseline and multi-perspective models

tokens since the majority of the documents (75%) fit this condition, and discarded the 54 extremely long documents. Before applying multi-perspective approach, 53 duplicate documents were eliminated as well. Regarding the labels, we removed all the documents annotated with *link not-working*, i.e. 54 documents, as they do not provide any information for the stance classification task. After removing these documents the dataset size became 1062, the original label distribution of the dataset is depicted in Figure 2a and the label distribution after the majority voting applied is shown in Figure 2b. For the fine-tuning, the dataset was splitted into train (80%), validation (20%), and test sets (20%). Note that the multi-perspective approach was applied only to the training and validation sets, while the same test set (after majority voting applied) was used for the model evaluation of the both multi-perspective and baseline approaches.

Apart from these, regarding the longer documents, we also experimented with LongFormer [18] that can handle longer inputs. Yet, due to its poor performance, we decided to discard it and proceed with chunking for both baseline and multi-perspective models. Note that the overlapping chunks contained 20 tokens.

4.2. Models overview

For our project, we decided to experiment with small language models, specifically with BERT-based models , as well as with larger models.

BERT which stands for Bidirectional Encoder Representations from Transformers [19] is a pre-trained language model developed by Google in 2017. The "Base" version of BERT consists of 12 layers (transformer blocks), 768 hidden units per layer, and 12 attention heads, resulting in a total of 110 million parameters. The "Uncased" version of BERT means that the text input is lowercased, and case information is not preserved.

BERT utilizes a bidirectional training approach, meaning that it considers both the left and right context when predicting a word, which is different from traditional models that typically process text either from left-to-right or right-to-left. This bidirectional nature allows BERT to gain a deeper understanding of language context and ambiguity.

The BERT model is pre-trained on two tasks:

- **Masked Language Model (MLM)**: Randomly masks some of the tokens in the input and aims to predict the original token based only on its context.
- **Next Sentence Prediction (NSP)**: Predicts whether a given pair of sentences are consecutive sentences in the original text.

These pre-training tasks enable BERT to learn contextual relationships between words (or sub-words) in a text corpus.

RoBERTa We experimented also with a bigger version of BERT, RoBERTa (A Robustly Optimized BERT Pretraining Approach) [20] is a variant of BERT developed by Facebook AI that enhances BERT's training methodology. The "Base" version of RoBERTa also consists of 12 layers, 768 hidden units per layer, and 12 attention heads, maintaining the same architecture and number of parameters as BERT-Base (110 million parameters).

RoBERTa improves upon BERT by making several key modifications to the pre-training process:

- **Removal of Next Sentence Prediction (NSP)**: RoBERTa eliminates the NSP task, which was found to be less beneficial for performance.
- **Training with Larger Batches and Learning Rates**: RoBERTa increases the batch size and learning rate, enabling more robust training.
- **Longer Training on More Data**: RoBERTa is trained on a significantly larger dataset (160GB of text) compared to BERT (16GB of text), and for a longer period.
- **Dynamic Masking**: Instead of static masking used in BERT, RoBERTa applies dynamic masking, where masking patterns are generated every time a sequence is fed into the model. These enhancements allow RoBERTa to achieve superior performance on many NLP benchmarks compared to the original BERT model.

Mistral We undertook an experiment with larger open-source models to assess potential performance differences, specifically focusing on fine-tuning Mistral and Llama 2. We fine-tuned Mistral 7B [21], a decoder-only model with 7.3 billion parameters, which, as of 2023, was the most advanced open-source model available. Mistral demonstrated strong reasoning skills and was rigorously evaluated using various benchmarks that tested its comprehension, reasoning, knowledge, and ability to handle diverse data types. It has been proven to be fluent in multiple languages, including English, French, Spanish, and German. Notably, Mistral 7B employs grouped-query attention (GQA), a variant of the standard attention mechanism. GQA optimizes the attention mechanism in transformer-based models by grouping queries together and computing their attention jointly, rather than independently. This approach reduces the number of attention computations, resulting in faster inference times.

Llama 2 Llama 2 is one of the biggest open source model [22], it largely adopts the pre-training settings and model architecture of Llama 1. It is trained on a vast dataset containing 2 trillion tokens from publicy available sources, employing a bytepair encoding (BPE) algorithm for tokenization. The model utilizes the standard transformer architecture, with pre-normalization using RMSNorm, the SwiGLU activation function, and rotary positional embedding. Notably, it supports an increased context length equal to 4096. Regarding hyperparameters, the model utilizes the AdamW optimizer and incorporates a cosine learning rate schedule with a warm-up period of 2000 steps. The final learning rate decays to 10Llama 2 demonstrates robust performance across various tasks, including coding, question answering in context, commonsense reasoning, and knowledge benchmarks.

4.3. Results

The model evaluation results on the test set after the fine-tuning for the encoder-based models of BERT-base and RoBERTa-base [19,20] are shown in Table 1. The fine-tuning has been fulfilled with a batch size of 32 for four epochs using the default hyperparameters on a 32GB Tesla V100 GPU. For the evaluation, we reported different metrics as accuracy, precision, recall and F1 score, while we compare the models based on the F1 since the dataset is slightly imbalanced towards *neutral*. Apart from these evaluation metrics, we also reported the model confidence scores which were computed with weighted average for models with chunking by taking into account the chunk length as well.

Based on the results, multi-perspective models consistently outperform the baseline models in all instances. Additionally, chunking works better than no chunking as expected, since with chunking we provide more information to the model. These findings apply to both BERT-base and RoBERTa-base, as well as to Llama 2 and Mistral. Notably, Llama 2 and Mistral do not require chunking due to their tolerance for a larger maximum input length than BERT. The best-performing baseline model is RoBERTa-base with chunking, whereas the best multi-perspective model is BERT-base with chunking, although the RoBERTA-base with chunking has a comparable performance (50.21 vs. 47.45) as well. This is probably due to the small dataset size or the effect of chunking.

Despite the slightly better performance of BERT over RoBERTa with chunking, RoBERTa is more confident in prediction and this applies to all the results in Table 1. Also, regardless of the model, it is more confident with chunking for both the baseline and multi-perspective models.

However, overall the most confident model both in baseline and multi-perspective is LLama 2 with a score of 0.66. We hypothesize that this confidence is misleading given the low scores in terms of accuracy or F1 score, indicating a case of high confidence but low accuracy despite the actual correct predictions.

5. Conclusion

In this work, we present a methodology to incorporate multi-perspective models into the stance detection task. Initially, we employed two transformer-based models, namely BERT-base and RoBERTa-base, among the first LLMs, for our experimental setup. We evaluated these models with and without chunking to address the sequence length limitation. Subsequently, we also experimented with larger models such as Mistral 7B and Llama 2 7B, which do not require chunking due to their larger input size.

Our initial studies demonstrate that multi-perspective models outperform the baseline models with and without chunking regardless of the model itself. Moreover, we observed that chunking improves model performance as well as the model confidence for both the baseline and multi-perspective settings. These results highlight that developing more inclusive perspective-aware AI models improves the classification model performance in subjective NLP tasks.

As a future direction, we aim to conduct hyperparameter tuning and explore data augmentation techniques. Specifically, we plan to augment the dataset by adding a document three times only if it has three different annotations, thus investigating various data augmentation strategies. We plan to have a more detailed analysis, e.g. query-wise, related to the annotator disagreement level on model performance as well as confidence scores. Finally, a similar methodology could be applied on *ideology detection* which is the second main task in the original paper of the dataset.

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